*9.55 Find **Z** in the network of Fig. 9.62, given that $V_o = 4\angle 0^\circ$ V.

ML

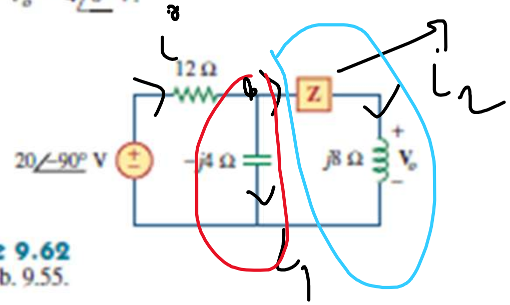


Figure 9.62
For Prob. 9.55.

$$i_2 = \frac{V_o}{j8} = \frac{4\angle 0^\circ}{j8}$$

$$= 0.5\angle -90^\circ$$

$$V_{-j4} = i_2 \times (Z + j8)$$

$$\therefore i_1 = \frac{V_{-j4}}{-j4} = \frac{i_2(Z+j8)}{-j4}$$

$$i = i_1 + i_2$$

$$= 0.5\angle{-90^\circ} + \frac{0.5\angle{-90^\circ}(Z + j8)}{-j4}$$

*9.55 Find **Z** in the network of Fig. 9.62, given that $\mathbf{V}_o = 4\angle{0^\circ}$ V.

ML

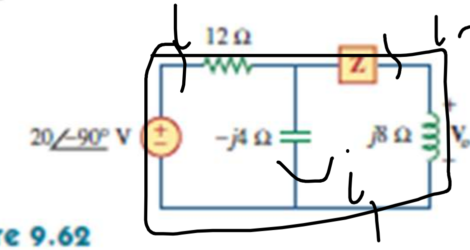


Figure 9.62
For Prob. 9.55.

KVL at outer loop,

*9.55 Find **Z** in the network of Fig. 9.62, given that $\mathbf{V}_o = 4\angle 0^\circ$ V.

ML

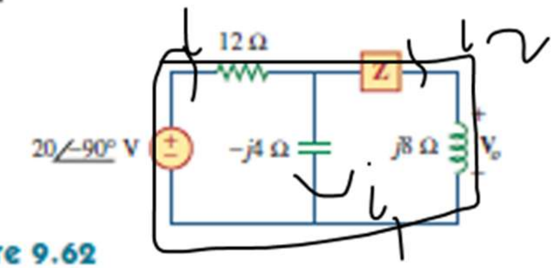


Figure 9.62
For Prob. 9.55.

$$-20\angle -90^\circ + i \times 12 + i_2(Z + j8) = 0$$

$$\Rightarrow \frac{(0.5\angle -90^\circ)(Z + j8)}{-j4} \times 12 + 0.5\angle -90^\circ (Z + j8) = 20\angle -90^\circ$$

$$\Rightarrow \frac{1}{8} \times (7 + j8) \times 12 + 0.5 \angle 90^\circ Z + 4$$

$$= 20 \angle 90^\circ$$

$$\Rightarrow 1.5Z + j12 + 0.5 \angle -90^\circ Z + 4$$

$$= 28 \angle -90^\circ$$

$$\Rightarrow Z\left(1.5 + 0.5\angle -90^\circ\right) = 20\angle -90^\circ$$

$$- 4 - j12$$

$$Z = 20.37\angle -78.7^\circ$$

9.66 For the circuit in Fig. 9.73, calculate $\mathbf{Z}_T$ and $\mathbf{V}_{ab}$.

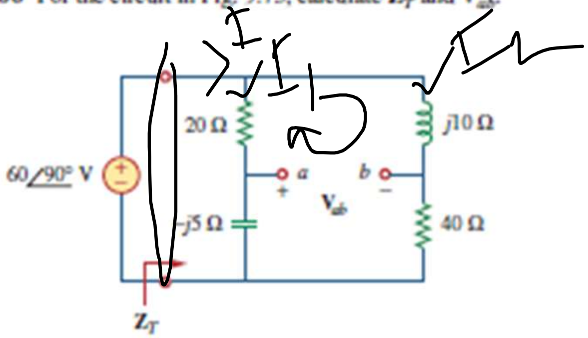


Figure 9.73
For Prob. 9.66.

$$Z_T = (20 - j5) \parallel (40 + j10)$$

$$= 14.12\angle -4.8^\circ$$

$$I = \frac{60\angle 90^\circ}{Z_T}$$

$$= 4.25\angle 94.8^\circ$$

9.66 For the circuit in Fig. 9.73, calculate $\mathbf{Z}_T$ and $\mathbf{V}_{ab}$.

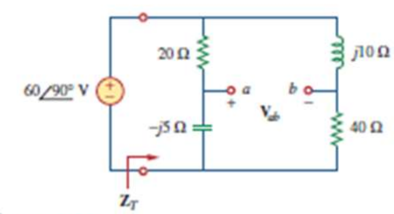


Figure 9.73
For Prob. 9.66.

$$I_1 = I \times \frac{40 + j10}{60 + 5j}$$

$$= 2.91 \angle 104.07^\circ$$

$$I_2 = \frac{20 - j5}{60 + 5j} \times I$$

$$= 1.45 \angle 76^\circ$$

9.66 For the circuit in Fig. 9.73, calculate $\mathbf{Z}_T$ and $\mathbf{V}_{ab}$.

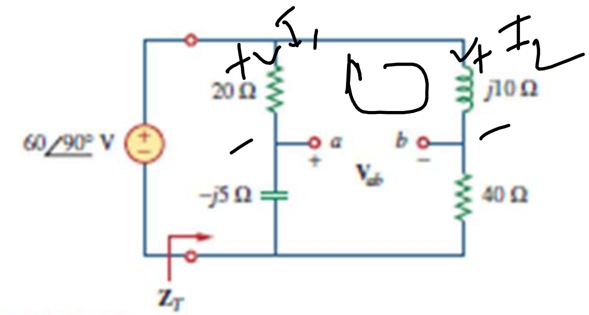


Figure 9.73
For Prob. 9.66.

KVL 1

$$-V_{ab} - 20 \times I_1 + j10 I_2 = 0$$

$$\Rightarrow V_{ab} = -20 I_1 - j10 \times I_2$$

$$=$$

9.63 For the circuit in Fig. 9.70, find the value of Z_T.

ML

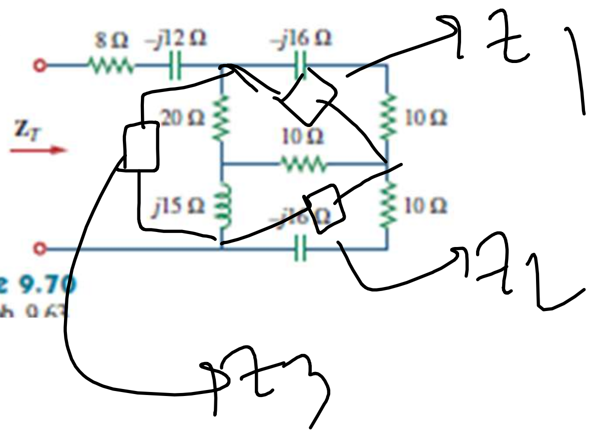


Figure 9.70
For Prob. 9.63

9.63 For the circuit in Fig. 9.70, find the value of Z_T.

ML

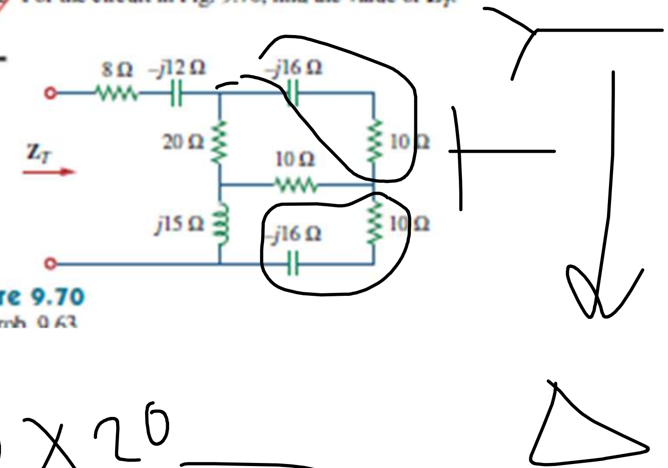


Figure 9.70
For Prob. 9.63

$$Z_1 = \frac{20 \times 10 + 10 \times j15 + j15 \times 20}{j15}$$

$$= 32.83 \angle -23.96^\circ$$

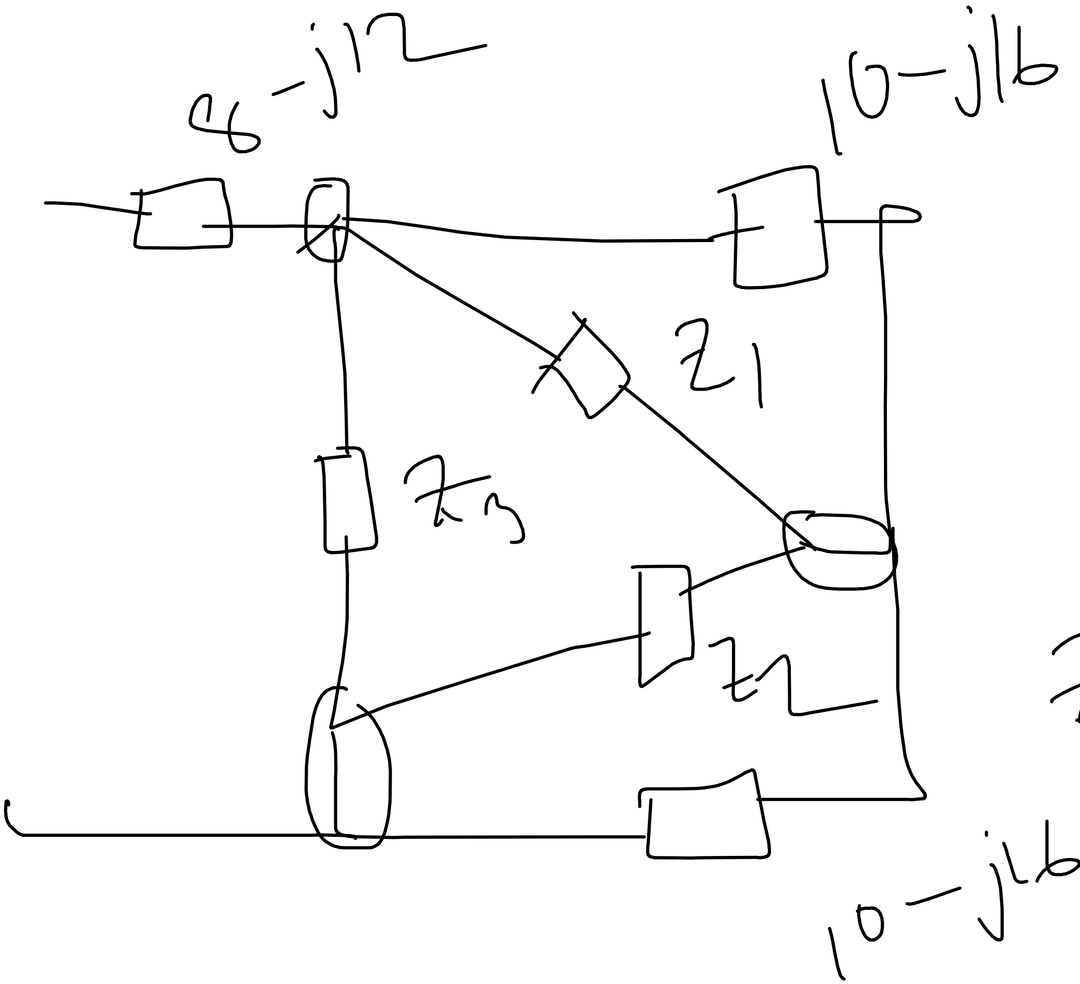
$200 + 450j$

$$z_2 = \frac{200 + 450j}{20}$$

$$= 24.62 \angle 66^{\circ}$$

$$z_3 = \frac{200 + 450j}{10}$$

$$= 49.24 \angle 66^{\circ}$$



9.63 For the circuit in Fig. 9.70, find the value of Z_T.

ML

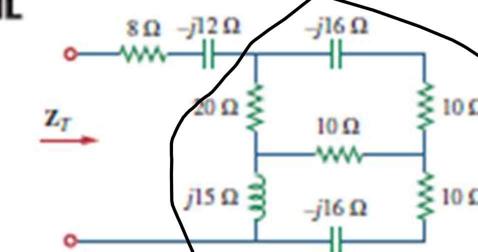
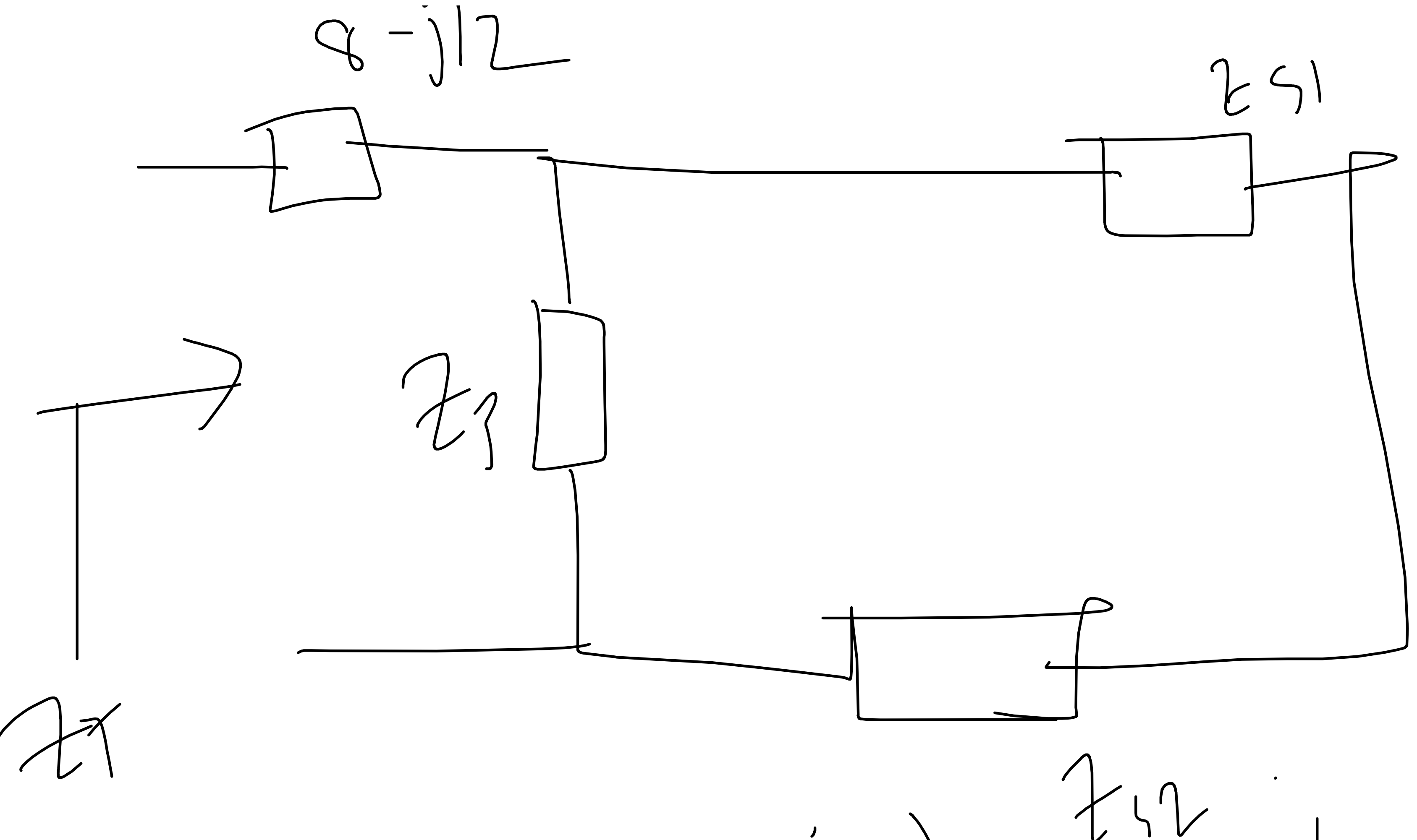


Figure 9.70
For Prob. 9.63

$$Z_{s1} = Z_1 \parallel (10 - j16)$$
$$= 12.48 \angle -45^\circ$$
$$Z_{s2} = Z_2 \parallel (10 - j16)$$
$$= 23.5 \angle -5.64$$



$$Z_T = (8 - j12) + Z_3 \,||\, (Z_{s1} + Z_{s2})$$

Nodal Analysis

Steps

(1) find nodes
1 node → ref node
rest are non-ref node

(2) Apply KCL to the non-ref node

10.8 Use nodal analysis to find current i_o in the circuit of Fig. 10.57. Let $i_s = 6\cos(200t + 15°)$ A.
ML

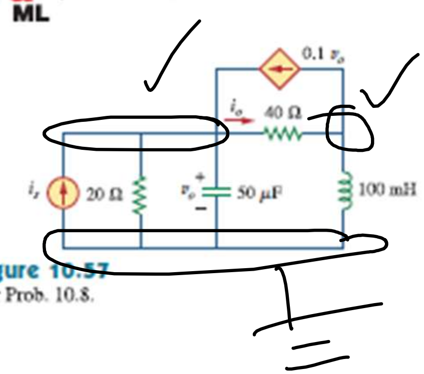


Figure 10.57
For Prob. 10.8.

③ Sol^n KCL eq^n ✓

10.8 Use nodal analysis to find current i_o in the circuit of Fig. 10.57. Let $i_s = 6\cos(200t + 15^\circ)$ A.

ML

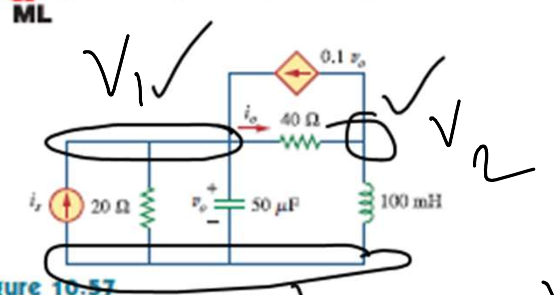


Figure 10.57
For Prob. 10.8.

KCL-1

$$50\mu F \rightarrow \frac{1}{j \times 200 \times 50\mu} = -j100$$

$$I_s = 6\angle 15^\circ$$

$$6\angle 15^\circ + 0.1 V_o = \frac{V_1}{20} + \frac{V_1}{-j100} + \frac{V_1 - V_2}{40}$$

$$V_o = V_1$$

$$6\angle 15^\circ = V_1\left(\frac{1}{20} - \frac{1}{j100} + \frac{1}{40} - 0.1\right)$$

$$-\frac{V_2}{40}$$

$$1$$

10.8 Use nodal analysis to find current i_o in the circuit of Fig. 10.57. Let $i_s = 6\cos(200t + 15°)$ A.

ML

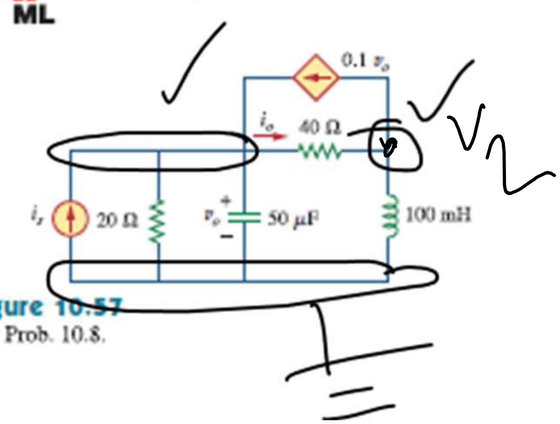


Figure 10.57
For Prob. 10.8.

KCL-2

100mH $\rightarrow j\omega(100m) = j20$

$$v_o = V_1$$

$$0.1V_1 + \frac{V_2}{j20} = \frac{V_1 - V_2}{40}$$

$$\Rightarrow V_1\left(0.1 - \frac{1}{40}\right) + V_2\left(\frac{1}{j20} + \frac{1}{40}\right) = 0 \quad \text{— } \boxed{2}$$

$$i_0 = \frac{V_1 - V_2}{40}$$

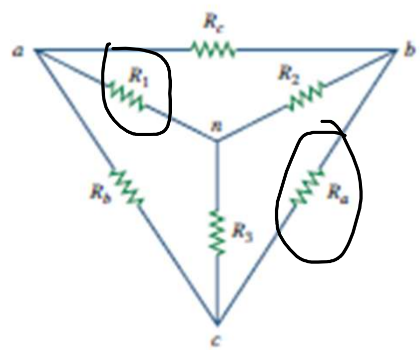
solving eqn 1 & 2,

$V_1 = 143.52 \angle -120^\circ$

$V_2 = 195.2 \angle 125^\circ$.

$$i_0 = \frac{V_1 - V_L}{40}$$

$$= 1.302 \angle -41^\circ$$



$$\boxed{R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}}$$

$$\boxed{R_1 = \frac{R_bR_c}{R_a + R_b + R_c}}$$